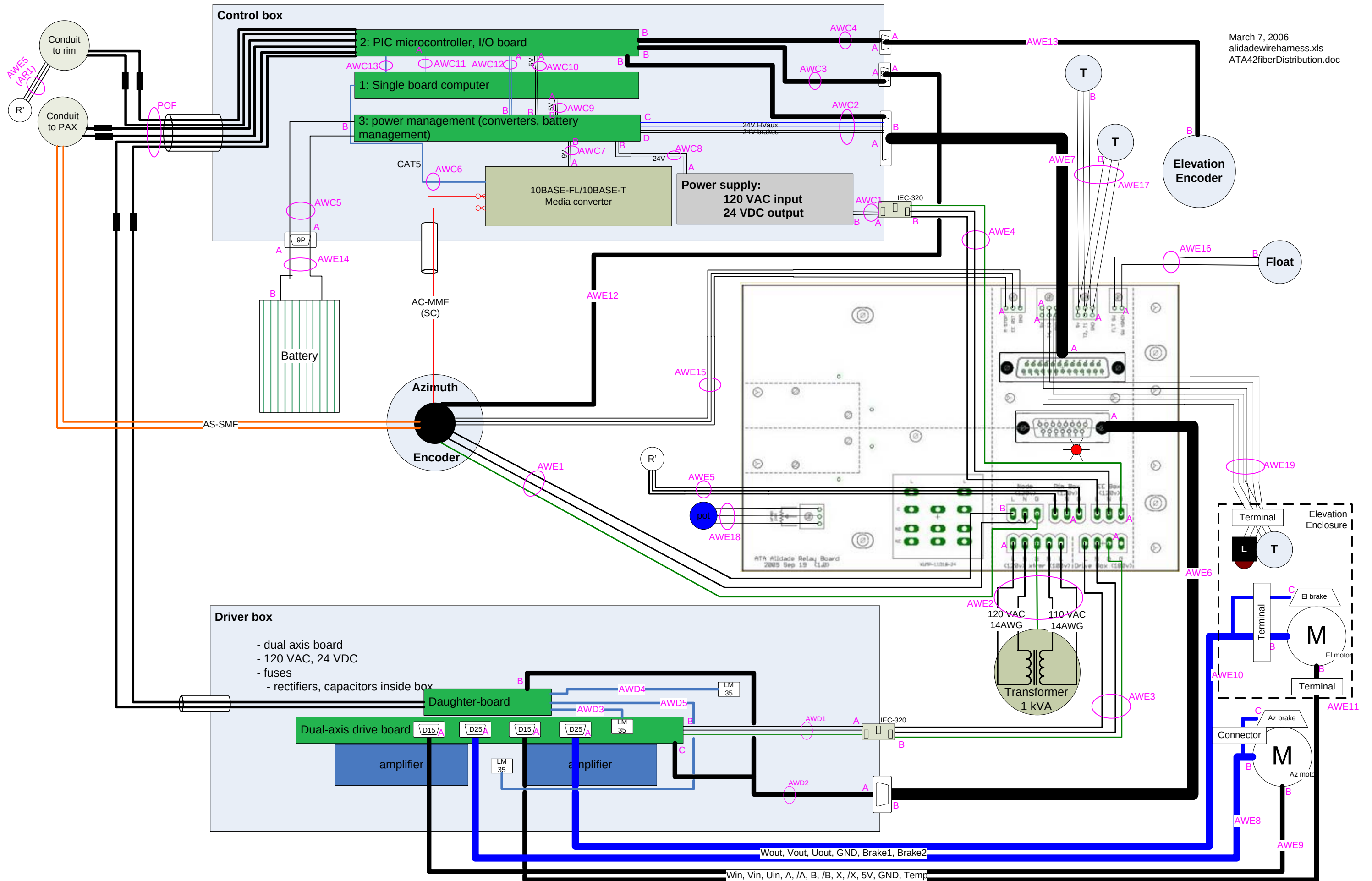
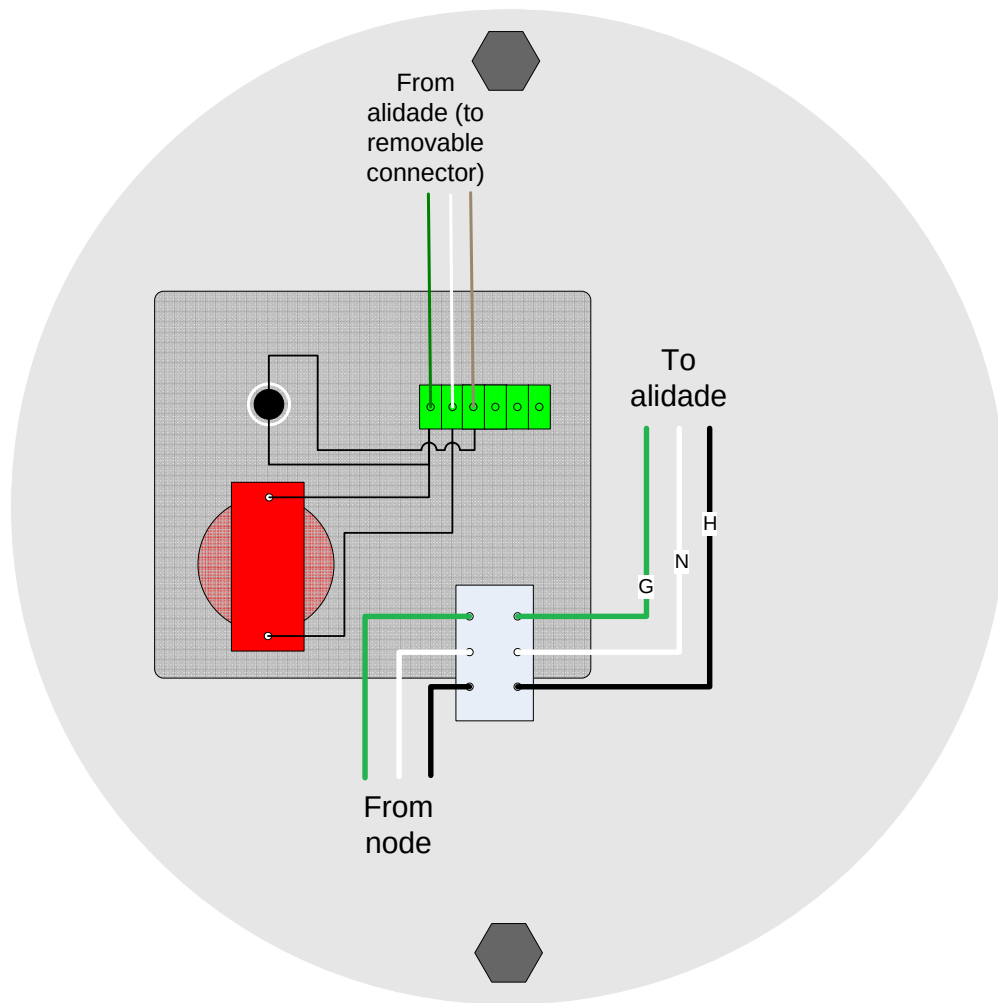
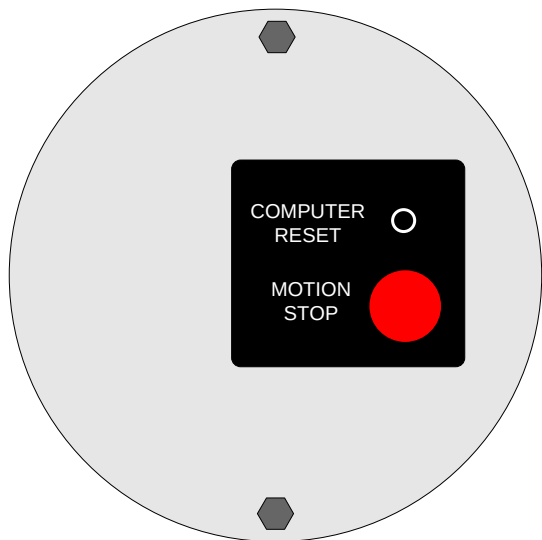




March 7, 2006  
alidawireharness.xls  
ATA42fiberDistribution.doc



| Alidade wire harnessing Enclosure                        |    |                 |                                                              |                                                                       |       |                   |       |        |
|----------------------------------------------------------|----|-----------------|--------------------------------------------------------------|-----------------------------------------------------------------------|-------|-------------------|-------|--------|
| Reference ELMCOOverview.vsd                              |    |                 |                                                              |                                                                       |       |                   |       |        |
| 7-Mar-06                                                 |    |                 |                                                              |                                                                       |       |                   |       |        |
| AWE1 Input power wires through pedestal                  |    |                 |                                                              |                                                                       |       |                   |       |        |
|                                                          | A  | Description     | B                                                            | Comment                                                               | Color | Size              | Twist | Length |
| Wires                                                    | 1  | 120VAC          |                                                              | 65 strand wire                                                        | white | #12               |       |        |
|                                                          | 2  | Return          |                                                              | Beldon 8527                                                           | black | #12               |       |        |
|                                                          | 3  | Ground          |                                                              | Beldon 8520                                                           | green | #14               |       |        |
| Ends                                                     | A  | bare wire       | screw terminal block behind pie-plate                        |                                                                       |       |                   |       |        |
|                                                          | B  | 39860-0103      | Molex terminal on Alidade Relay Board                        |                                                                       |       |                   |       |        |
| AWE2 Transformer                                         |    |                 |                                                              |                                                                       |       |                   |       |        |
|                                                          | A  | Description     |                                                              | Comment                                                               | Color | Size              | Twist | Length |
| Wires                                                    | 1  | 120VAC          |                                                              | These wires except Ground are all on the toroid, we provide connector | white | #14               |       |        |
|                                                          | 2  | Return          |                                                              |                                                                       | black | #14               |       |        |
|                                                          | 3  |                 |                                                              |                                                                       |       |                   |       |        |
|                                                          | 4  | 100VAC          |                                                              |                                                                       | red   | #14               |       |        |
|                                                          | 5  | Return          |                                                              |                                                                       | red   | #14               |       |        |
| End                                                      | A  | 39860-0105      | Molex connector on alidade relay card                        |                                                                       |       |                   |       |        |
| AWE3 Power to motors                                     |    |                 |                                                              |                                                                       |       |                   |       |        |
|                                                          | A  | Description     | B                                                            | Comment                                                               | Color | Size              | Twist | Length |
| Wires                                                    | 1  | 110VAC          |                                                              | Use standard power cord and chop off plug                             | black | #14               |       | 2 ft   |
|                                                          | 2  | Return          |                                                              |                                                                       | white | #14               |       |        |
|                                                          | 3  | Ground          |                                                              |                                                                       | green | #14               |       |        |
| Ends                                                     | A  | 39860-0104      | Molex connector on alidade relay board                       |                                                                       |       |                   |       |        |
|                                                          | B  | IEC-320         | right angle plug on drive box                                |                                                                       |       |                   |       |        |
| AWE4 Power for control box                               |    |                 |                                                              |                                                                       |       |                   |       |        |
|                                                          | A  | Description     | B                                                            | Comment                                                               | Color | Size              | Twist | Length |
| Wires                                                    | 1  | 120VAC          |                                                              | Use standard power cord and chop off plug                             | white | #18               |       |        |
|                                                          | 2  | Return          |                                                              |                                                                       | black | #18               |       |        |
|                                                          | 3  | Ground          |                                                              |                                                                       | green | #18               |       |        |
| Ends                                                     | A  | 39860-0103      | Molex connector on alidade relay board                       |                                                                       |       |                   |       |        |
|                                                          | B  | IEC-320         | right angle plug on control box (may not exist in rt angle?) |                                                                       |       |                   |       |        |
| AWE5 Power for rim power box (also called AR1)           |    |                 |                                                              |                                                                       |       |                   |       |        |
|                                                          | A  | Description     | B                                                            | Comment                                                               | Color | Size              | Twist | Length |
| Wires                                                    | 1  | 120VAC          |                                                              | wire                                                                  | black | #14               |       |        |
|                                                          | 2  | Return          |                                                              |                                                                       | white | #14               |       |        |
|                                                          | 3  | Ground          |                                                              |                                                                       | green | #14               |       |        |
| Ends                                                     | A  | 39860-0103      | Molex connector on alidade relay board                       |                                                                       |       |                   |       |        |
|                                                          | B  | ParlBlade U Gnd | connector in rim power box                                   |                                                                       |       |                   |       |        |
| AWE6 15-wire cable from alidade relay board to drive box |    |                 |                                                              |                                                                       |       |                   |       |        |
|                                                          | A  | Description     | B                                                            | Comment                                                               |       | Size              |       | Length |
| Wires/pins                                               | 1  | az brakes +24V  | 1                                                            | 1&9 are controlled with the main drives transformer power             |       |                   |       |        |
|                                                          | 2  | Ground          | 2                                                            |                                                                       |       |                   |       |        |
|                                                          | 3  | 24V aux         | 3                                                            |                                                                       |       | all #24           |       | 23"    |
|                                                          | 4  | Ground          | 4                                                            |                                                                       |       |                   |       |        |
|                                                          | 5  | drive enable +  | 5                                                            | 5&13 are connected through relay contacts which are closed            |       |                   |       |        |
|                                                          | 6  | Ground          | 6                                                            | when the main drives transformer power relay is latched               |       |                   |       |        |
|                                                          | 7  | Az cw limit     | 7                                                            | 7&15 are connected to 8 via NC switches                               |       |                   |       |        |
|                                                          | 8  | Az limit common | 8                                                            |                                                                       |       |                   |       |        |
|                                                          | 9  | el brakes +24V  | 9                                                            |                                                                       |       |                   |       |        |
|                                                          | 10 | Ground          | 10                                                           | Belden 9947                                                           | maybe |                   |       |        |
|                                                          | 11 | 24V aux         | 11                                                           | awaiting quote from cablestogo.com                                    |       |                   |       |        |
|                                                          | 12 | Ground          | 12                                                           |                                                                       |       |                   |       |        |
|                                                          | 13 | drive enable -  | 13                                                           |                                                                       |       |                   |       |        |
|                                                          | 14 | Ground          | 14                                                           |                                                                       |       |                   |       |        |
|                                                          | 15 | Az ccw limit    | 15                                                           |                                                                       |       |                   |       |        |
| Ends                                                     | A  | D15P            | Dsub to alidade relay board                                  |                                                                       |       |                   |       |        |
|                                                          | B  | D15S            | Dsub to drive box                                            |                                                                       |       | connects to AWD2A |       |        |

| AWE7 25-wire cable from alidade relay board to control box |    |                           |                             |                                                                   |                     |        |       |        |
|------------------------------------------------------------|----|---------------------------|-----------------------------|-------------------------------------------------------------------|---------------------|--------|-------|--------|
|                                                            | A  | Description               | B                           | Comment                                                           |                     |        |       | Length |
| Wires/pins                                                 | 1  | brakes +24V               | 1                           | 1,3,5,11,14,16,18, 24 are power and control signals from the      |                     |        |       |        |
|                                                            | 2  | Ground                    | 2                           | control box                                                       |                     |        |       |        |
|                                                            | 3  | 24V aux                   | 3                           |                                                                   |                     |        |       |        |
|                                                            | 4  | Ground                    | 4                           |                                                                   |                     |        |       |        |
|                                                            | 5  | 5V sensor +               | 5                           |                                                                   |                     |        |       |        |
|                                                            | 6  | Ground                    | 6                           |                                                                   |                     |        |       |        |
|                                                            | 7  | float in                  | 7                           | 7,8,9,10,12,13,20,21,22,23,25 are logic and sensor signals        |                     |        |       |        |
|                                                            | 8  | T input1 -                | 8                           | returning to the control box                                      |                     |        |       |        |
|                                                            | 9  | T input2 -                | 9                           |                                                                   |                     |        |       |        |
|                                                            | 10 | Az brake sense            | 10                          |                                                                   |                     |        |       |        |
|                                                            | 11 | Latch                     | 11                          | main drives transformer power is latched by holding 24 logic high |                     |        |       |        |
|                                                            | 12 | Pot 1                     | 12                          | while pulsing 11 logic low-high-low                               |                     |        |       |        |
|                                                            | 13 | Pot wiper                 | 13                          | wrap pot is wired to relay board such that resistance between 13  |                     |        |       |        |
|                                                            | 14 | brakes +24V               | 14                          | and 25 increases when azimuth rotates cw                          |                     |        |       |        |
|                                                            | 15 | Ground                    | 15                          |                                                                   |                     |        |       |        |
|                                                            | 16 | 24V aux                   | 16                          |                                                                   |                     |        |       |        |
|                                                            | 17 | Ground                    | 17                          |                                                                   |                     |        |       |        |
|                                                            | 18 | 5V sensor -               | 18                          | awaiting quote from cablestogo.com                                |                     |        |       |        |
|                                                            | 19 | Ground                    | 19                          |                                                                   |                     |        |       |        |
|                                                            | 20 | T input1 +                | 20                          |                                                                   |                     |        |       |        |
|                                                            | 21 | T input2 +                | 21                          |                                                                   |                     |        |       |        |
|                                                            | 22 | electronics reset         | 22                          |                                                                   |                     |        |       |        |
|                                                            | 23 | EI brake sense            | 23                          |                                                                   |                     |        |       |        |
|                                                            | 24 | Unlatch                   | 24                          |                                                                   |                     |        |       |        |
|                                                            | 25 | Pot 2                     | 25                          |                                                                   |                     |        |       |        |
| Ends                                                       | A  | D25S                      | Dsub to alidade relay board |                                                                   |                     |        |       |        |
|                                                            | B  | D25P                      | Dsub to control box         |                                                                   |                     |        |       |        |
| AWE8 motor power and brakes for azimuth                    |    |                           |                             |                                                                   |                     |        |       |        |
|                                                            | A  | Description               | B                           | Comment                                                           | Color               | Size   | Twist | Length |
| Wires/pins                                                 | 1  | W1                        | W                           |                                                                   | purple              | #14    |       |        |
|                                                            | 2  | W2                        |                             |                                                                   |                     |        |       |        |
|                                                            | 14 | W3                        |                             |                                                                   |                     |        |       |        |
|                                                            | 15 | W4                        |                             |                                                                   |                     |        |       |        |
|                                                            | 3  | N/C                       |                             |                                                                   |                     |        |       |        |
|                                                            | 16 | N/C                       |                             |                                                                   |                     |        |       |        |
|                                                            | 4  | V1                        | V                           |                                                                   | brown               | #14    |       |        |
|                                                            | 5  | V2                        |                             |                                                                   |                     |        |       |        |
|                                                            | 17 | V3                        |                             |                                                                   |                     |        |       |        |
|                                                            | 18 | V4                        |                             |                                                                   |                     |        |       |        |
|                                                            | 6  | N/C                       |                             |                                                                   |                     |        |       |        |
|                                                            | 19 | N/C                       |                             |                                                                   |                     |        |       |        |
|                                                            | 7  | U1                        | U                           |                                                                   | blue                | #14    |       |        |
|                                                            | 8  | U1                        |                             |                                                                   |                     |        |       |        |
|                                                            | 20 | U1                        |                             |                                                                   |                     |        |       |        |
|                                                            | 21 | U1                        |                             |                                                                   |                     |        |       |        |
|                                                            | 9  | N/C                       |                             |                                                                   |                     |        |       |        |
|                                                            | 22 | N/C                       |                             |                                                                   |                     |        |       |        |
|                                                            | 10 | G1                        | Ground                      |                                                                   | green               | #14    |       |        |
|                                                            | 11 | G2                        |                             |                                                                   |                     |        |       |        |
|                                                            | 23 | G3                        |                             |                                                                   |                     |        |       |        |
|                                                            | 24 | G4                        |                             |                                                                   |                     |        |       |        |
|                                                            | 12 | BP                        |                             |                                                                   | blue                | #16    |       |        |
|                                                            | 13 | BM                        |                             |                                                                   | blue                | #16    |       |        |
|                                                            | 25 | SHIELD                    |                             |                                                                   | shield              | shield |       |        |
| Ends                                                       | A  | D25P                      | filtered D25S on board      |                                                                   |                     |        |       |        |
|                                                            | B  | 39-01-2063-P/39-00-0041-C |                             |                                                                   |                     |        |       |        |
|                                                            | C  | 39-01-3023-P/39-00-0041-C |                             |                                                                   | wires part of brake |        |       |        |

| AWE9 encoders and hall's for azimuth       |    |                    |        |                          |             |        |       |        |
|--------------------------------------------|----|--------------------|--------|--------------------------|-------------|--------|-------|--------|
|                                            | A  | Description        | B      | Comment                  | Color       | Size   | Twist | Length |
| Wires/pins                                 | 1  | N/C                |        | GND on board             | N/C         |        |       |        |
|                                            | 2  | U                  |        |                          | brown       |        |       |        |
|                                            | 3  | V                  |        |                          | gray        |        |       |        |
|                                            | 4  | W                  |        |                          | white       |        |       |        |
|                                            | 5  | GND                |        |                          | black       |        | V     |        |
|                                            | 6  | X                  |        |                          | purple      |        | Z     |        |
|                                            | 7  | B                  |        |                          | green       |        | Y     |        |
|                                            | 8  | A                  |        |                          | blue        |        | X     |        |
|                                            | 9  | thermal sensor     |        | GPIN5                    | orange      |        | W     |        |
|                                            | 10 | GND thermal sensor |        | GND                      | orang/white |        | W     |        |
|                                            | 11 | Vcc (+5V)          |        |                          | red         |        | V     |        |
|                                            | 12 | N/C                |        | GND on board             | N/C         |        |       |        |
|                                            | 13 | /X                 |        |                          | purple/blk  |        | Z     |        |
|                                            | 14 | /B                 |        |                          | green/blk   |        | Y     |        |
|                                            | 15 | /A                 |        |                          | blue/blk    |        | X     |        |
| Ends                                       | A  | D15P               |        | filtered D15S on board   |             |        |       |        |
|                                            | B  | M                  |        |                          |             |        |       |        |
| AWE10 motor power and brakes for elevation |    |                    |        |                          |             |        |       |        |
|                                            | A  | Description        | B      | Comment                  | Color       | Size   | Twist | Length |
| Wires/pins                                 | 1  | W1                 | W      |                          | purple      | #14    |       |        |
|                                            | 2  | W2                 |        |                          |             |        |       |        |
|                                            | 14 | W3                 |        |                          |             |        |       |        |
|                                            | 15 | W4                 |        |                          |             |        |       |        |
|                                            | 3  | N/C                |        |                          |             |        |       |        |
|                                            | 16 | N/C                |        |                          |             |        |       |        |
|                                            | 4  | V1                 | V      |                          | brown       | #14    |       |        |
|                                            | 5  | V2                 |        |                          |             |        |       |        |
|                                            | 17 | V3                 |        |                          |             |        |       |        |
|                                            | 18 | V4                 |        |                          |             |        |       |        |
|                                            | 6  | N/C                |        |                          |             |        |       |        |
|                                            | 19 | N/C                |        |                          |             |        |       |        |
|                                            | 7  | U1                 | U      |                          | blue        | #14    |       |        |
|                                            | 8  | U1                 |        |                          |             |        |       |        |
|                                            | 20 | U1                 |        |                          |             |        |       |        |
|                                            | 21 | U1                 |        |                          |             |        |       |        |
|                                            | 9  | N/C                |        |                          |             |        |       |        |
|                                            | 22 | N/C                |        |                          |             |        |       |        |
|                                            | 10 | G1                 | Ground |                          | green       | #14    |       |        |
|                                            | 11 | G2                 |        |                          |             |        |       |        |
|                                            | 23 | G3                 |        |                          |             |        |       |        |
|                                            | 24 | G4                 |        |                          |             |        |       |        |
|                                            | 12 | BP                 |        |                          | blue        | #16    |       |        |
|                                            | 13 | BM                 |        |                          | blue        | #16    |       |        |
|                                            | 25 | SHIELD             |        |                          | shield      | shield |       |        |
| Ends                                       | A  | D25P               |        | plugs into D25S on board |             |        |       |        |
|                                            | B  | 39-01-2063-P       | Molex  |                          |             |        |       |        |
|                                            | C  | 39-01-3023-P       | Molex  |                          |             |        |       |        |

| AWE11 encoders and hall's for elevation                        |    |                    |   |                        |             |      |       |        |
|----------------------------------------------------------------|----|--------------------|---|------------------------|-------------|------|-------|--------|
|                                                                | A  | Description        | B | Comment                | Color       | Size | Twist | Length |
| Wires/pins                                                     | 1  | N/C                |   | GND on board           | N/C         |      |       |        |
|                                                                | 2  | U                  |   |                        | brown       |      |       |        |
|                                                                | 3  | V                  |   |                        | gray        |      |       |        |
|                                                                | 4  | W                  |   |                        | white       |      |       |        |
|                                                                | 5  | GND                |   |                        | black       |      | V     |        |
|                                                                | 6  | X                  |   |                        | purple      |      | Z     |        |
|                                                                | 7  | B                  |   |                        | green       |      | Y     |        |
|                                                                | 8  | A                  |   |                        | blue        |      | X     |        |
|                                                                | 9  | thermal sensor     |   | GPIN5                  | orange      |      | W     |        |
|                                                                | 10 | GND thermal sensor |   | GND                    | orang/white |      | W     |        |
|                                                                | 11 | Vcc (+5V)          |   |                        | red         |      | V     |        |
|                                                                | 12 | N/C                |   | GND on board           | N/C         |      |       |        |
|                                                                | 13 | /X                 |   |                        | purple/blk  |      | Z     |        |
|                                                                | 14 | /B                 |   |                        | green/blk   |      | Y     |        |
|                                                                | 15 | /A                 |   |                        | blue/blk    |      | X     |        |
| Ends                                                           | A  | D15P               |   | filtered D15S on board |             |      |       |        |
|                                                                | B  | D15P               |   | D 15p on motor         |             |      |       |        |
| AWE12 encoder for azimuth (P/N: R176H-03600Q-5L10-AT40SP-24MN) |    |                    |   |                        |             |      |       |        |
|                                                                | A  | Description        | B | Comment                | Color       | Size | Twist | Length |
| Wires/pins                                                     | 1  | A                  |   |                        | yellow      |      |       |        |
|                                                                | 2  | B                  |   |                        | green       |      |       |        |
|                                                                | 3  | IND                |   |                        | blue        |      |       |        |
|                                                                | 4  | N/C                |   |                        | shield      |      |       |        |
|                                                                | 5  | +V                 |   |                        | red         |      |       |        |
|                                                                | 6  | /A                 |   |                        | brown       |      |       |        |
|                                                                | 7  | /B                 |   |                        | orange      |      |       |        |
|                                                                | 8  | /IND               |   |                        | white       |      |       |        |
|                                                                | 9  | COMMON             |   |                        | black       |      |       |        |
| Ends                                                           | A  | D9P                |   | Dsub on control box    |             |      |       |        |
|                                                                | B  | --                 |   | cable part of encoder  |             |      |       |        |
| AWE13 encoder for elevation (P/N: 9220S03600D5L10D99SP04EA)    |    |                    |   |                        |             |      |       |        |
|                                                                | A  | Description        | B | Comment                | Color       | Size | Twist | Length |
| Wires/pins                                                     | 1  | A                  |   |                        | yellow      |      |       |        |
|                                                                | 2  | B                  |   |                        | green       |      |       |        |
|                                                                | 3  | IND                |   |                        | blue        |      |       |        |
|                                                                | 4  | N/C                |   |                        | shield      |      |       |        |
|                                                                | 5  | +5V                |   |                        | red         |      |       |        |
|                                                                | 6  | /A                 |   |                        | brown       |      |       |        |
|                                                                | 7  | /B                 |   |                        | orange      |      |       |        |
|                                                                | 8  | /IND               |   |                        | white       |      |       |        |
|                                                                | 9  | COMMON             |   |                        | black       |      |       |        |
| Ends                                                           | A  | D9P                |   | Dsub on control box    |             |      |       |        |
|                                                                | B  | --                 |   | cable part of encoder  |             |      |       |        |
| AWE14 battery for control box                                  |    |                    |   |                        |             |      |       |        |
|                                                                | A  | Description        | B | Comment                | Color       | Size | Twist | Length |
| Wires/pins                                                     | 1  | +12V               | 1 | Only 2 wires           | red         |      |       |        |
|                                                                | 2  | +12V               | 1 |                        | red         |      |       |        |
|                                                                | 3  | N/C                |   |                        |             |      |       |        |
|                                                                | 4  | GND                | 2 |                        | black       |      |       |        |
|                                                                | 5  | GND                | 2 |                        | black       |      |       |        |
|                                                                | 6  | +12V               | 1 |                        | red         |      |       |        |
|                                                                | 7  | N/C                |   |                        |             |      |       |        |
|                                                                | 8  | N/C                |   |                        |             |      |       |        |
|                                                                | 9  | GND                | 2 |                        | black       |      |       |        |
| Ends                                                           | A  | D9S                |   |                        |             |      |       |        |
|                                                                | B  | Spade              |   |                        |             |      |       |        |

| AWE15 Signal from relay board to pie-plate LappUSA (Olflex) 900P Cable, 3 conductor #24, 811442 |   |                               |              |                           |                        |      |       |
|-------------------------------------------------------------------------------------------------|---|-------------------------------|--------------|---------------------------|------------------------|------|-------|
|                                                                                                 | A | Description                   | B            | Comment                   | Color                  | Size | Twist |
| Wires                                                                                           | 1 | common                        | 1            |                           | green                  | #24  |       |
|                                                                                                 | 2 | M-stop                        | 2            |                           | white                  | #24  |       |
|                                                                                                 | 3 | reset                         | 3            |                           | brown                  | #24  |       |
| Ends                                                                                            | A | Molex 43645-0300/43030-009    |              | on alidade relay board    | (Jameco 338360/337738) |      |       |
|                                                                                                 | B | bare                          |              | terminal behind pie plate |                        |      |       |
| AWE16 signal from oil float                                                                     |   |                               |              |                           |                        |      |       |
|                                                                                                 | A | Description                   | B            | Comment                   | Color                  | Size | Twist |
| Wire                                                                                            | 1 |                               |              |                           |                        | #24  |       |
|                                                                                                 | 2 |                               |              |                           |                        |      |       |
| Ends                                                                                            | A | Molex 43645-0200/43030-009    |              | on alidade relay board    | (Jameco 338351/337738) |      |       |
|                                                                                                 | B | Crimps?                       |              |                           |                        |      |       |
| AWE17 Signal from temperature sensor to relay board                                             |   |                               |              |                           |                        |      |       |
|                                                                                                 | A | Description                   | B, C         | Comment                   | Color                  | Size | Twist |
| Wires                                                                                           | 1 | GND                           | B1           |                           | Black                  | #24  |       |
|                                                                                                 | 2 | T                             | B2           |                           | Yellow                 | #24  |       |
|                                                                                                 | 3 | +5V                           | B3           |                           | Red                    | #24  |       |
|                                                                                                 | 4 | GND                           | C1           |                           | Black                  | #24  |       |
|                                                                                                 | 5 | T                             | C2           |                           | Yellow                 | #24  |       |
|                                                                                                 | 6 | +5V                           | C3           |                           | Red                    | #24  |       |
| Ends                                                                                            | A | Molex 43025-0600/43030-009    |              | to alidade relay board    |                        |      |       |
|                                                                                                 | B | solder to LM35                |              | Lm-35 Device in TO-220    |                        |      |       |
|                                                                                                 | C | solder to LM35                |              | Lm-35 Device in TO-220    |                        |      |       |
| AWE18 Signal from az wrap potentiometer                                                         |   |                               |              |                           |                        |      |       |
|                                                                                                 | A | Description                   | B            | Comment                   | Color                  | Size | Twist |
| Wires                                                                                           | 1 |                               |              |                           | red                    | #24  |       |
|                                                                                                 | 2 |                               |              |                           | yellow                 |      |       |
|                                                                                                 | 3 |                               |              |                           | black                  |      |       |
| Ends                                                                                            | A | bare wire                     | posts on pot |                           |                        |      |       |
|                                                                                                 | B | Molex 43645-0300/43030-009    |              | on alidade relay board    |                        |      |       |
| AWE19 Signal from elevation level sensor and thermal sensor                                     |   |                               |              |                           |                        |      |       |
|                                                                                                 | A | Description                   | B            | Comment                   | Color                  | Size | Twist |
| Wires                                                                                           | 1 | red                           | +5V          |                           |                        | #24  |       |
|                                                                                                 | 2 |                               |              |                           |                        |      |       |
|                                                                                                 | 3 |                               |              |                           |                        |      |       |
|                                                                                                 | 4 |                               |              |                           |                        |      |       |
|                                                                                                 | 5 | green                         | signal       |                           |                        |      |       |
|                                                                                                 | 6 | blue                          | ground       |                           |                        |      |       |
| Ends                                                                                            | A | Molex 43025-0600/43030-009    |              | on relay board            |                        |      |       |
|                                                                                                 | B | solder/leads in elevation can |              |                           |                        |      |       |

|                                                               |                                              |                            |                                         |                       |              |                         |                    |         |         |         |
|---------------------------------------------------------------|----------------------------------------------|----------------------------|-----------------------------------------|-----------------------|--------------|-------------------------|--------------------|---------|---------|---------|
| Alidade wire harnessing: controller box                       |                                              |                            |                                         |                       |              |                         |                    |         |         |         |
| Reference ELMCOOverview.vsd                                   |                                              |                            |                                         |                       |              |                         |                    |         |         |         |
| 7-Mar-06                                                      |                                              |                            |                                         |                       |              |                         |                    |         |         |         |
| AWC1 Power from relay board to power mgmt board               |                                              |                            |                                         |                       |              |                         |                    |         |         |         |
|                                                               | A                                            | Description                | B                                       | Color                 | Size         | Twist                   | Length             |         |         |         |
| Wires                                                         | 1                                            | 100 VAC "L"                |                                         | White                 | 18           |                         | 15"                |         |         |         |
|                                                               | 2                                            | Return "N"                 |                                         | Black                 | 18           |                         | 15"                |         |         |         |
|                                                               | 3                                            | Ground                     |                                         | Green                 | 18           |                         | 15"                |         |         |         |
| Ends                                                          | A IEC-320 0.25 quick disconnect F insulated  |                            |                                         |                       |              |                         |                    |         |         |         |
|                                                               | B spade fork term #6, unins bent heat shrink |                            |                                         |                       |              |                         |                    |         |         |         |
| AWC2 25-wire cable from alidade relay board to control box    |                                              |                            |                                         |                       |              |                         |                    |         |         |         |
|                                                               | A                                            | Description                | B,C,D                                   | Comment               | Color        | Size                    | Twist              | Length  |         |         |
| Wires/pins                                                    | 1                                            | brakes +24V                | C1                                      | power mgmt board      | brown        | all wires #22 AWG       | Z                  | 7.5"    |         |         |
|                                                               | 2                                            | Ground                     | C2                                      |                       | black        |                         | Z                  | 7.5"    |         |         |
|                                                               | 3                                            | 24V aux                    | D1                                      |                       | purple       |                         | Y                  | 7.5"    |         |         |
|                                                               | 4                                            | Ground                     | D2                                      |                       | black        |                         | Y                  | 7.5"    |         |         |
|                                                               | 5                                            | +5V                        | B19                                     | microcontroller board | red          |                         | X                  | 7.5"    |         |         |
|                                                               | 6                                            | Ground                     | B15                                     |                       | black        |                         | X                  | 7.5"    |         |         |
|                                                               | 7                                            | float in                   | B6                                      |                       | green        |                         |                    | 7.5"    |         |         |
|                                                               | 8                                            | T 2 input                  | B9                                      |                       | blue         |                         |                    | 7.5"    |         |         |
|                                                               | 9                                            | T 4 input                  | B10                                     |                       | yellow       |                         |                    | 7.5"    |         |         |
|                                                               | 10                                           | Az brake sense             | B13                                     |                       | white        |                         |                    | 7.5"    |         |         |
|                                                               | 11                                           | Latch                      | B7                                      |                       | orange       |                         |                    | 7.5"    |         |         |
|                                                               | 12                                           | Pot 1 (GND)                | B18                                     |                       | black        |                         | W                  | 7.5"    |         |         |
|                                                               | 13                                           | Pot wiper                  | B16                                     |                       | green        |                         |                    | 7.5"    |         |         |
|                                                               | 14                                           | brakes +24V                | C1                                      | power mgmt board      | brown        |                         | Z                  | 7.5"    |         |         |
|                                                               | 15                                           | Ground                     | C2                                      |                       | black        |                         | Z                  | 7.5"    |         |         |
|                                                               | 16                                           | 24V aux                    | D1                                      |                       | purple       |                         | Y                  | 7.5"    |         |         |
|                                                               | 17                                           | Ground                     | D2                                      |                       | black        |                         | Y                  | 7.5"    |         |         |
|                                                               | 18                                           | +5V                        | B20                                     | microcontroller board | red          |                         | X                  | 7.5"    |         |         |
|                                                               | 19                                           | Ground                     | B3                                      |                       | black        |                         | X                  | 7.5"    |         |         |
|                                                               | 20                                           | T 1 input                  | B11                                     |                       | green        |                         |                    | 7.5"    |         |         |
|                                                               | 21                                           | T 3 input                  | B12                                     |                       | blue         |                         |                    | 7.5"    |         |         |
|                                                               | 22                                           | electronics reset (on/off) | B5                                      |                       | yellow       |                         |                    | 7.5"    |         |         |
|                                                               | 23                                           | EI brake sense             | B8                                      |                       | white        |                         |                    | 7.5"    |         |         |
|                                                               | 24                                           | Unlatch (drives on/off)    | B14                                     |                       | grey         |                         |                    | 7.5"    |         |         |
|                                                               | 25                                           | Pot 2 (+5V)                | B17                                     |                       | red          |                         | W                  | 7.5"    |         |         |
| Ends                                                          | A 56-126-004                                 |                            | filtered D25 Socket on top of control b |                       |              | connects to AWE7B       |                    |         |         |         |
|                                                               | B WM2527-ND/WM62510-ND                       |                            | 2x10 100mil ucntlr board                |                       |              | (22-55-2201/16-02-0102) |                    |         |         |         |
|                                                               | C 770849-2/770522-1                          |                            | power mgmt board                        |                       | relay/brakes | JameCo 236321           |                    |         |         |         |
|                                                               | D 770849-2/770522-1                          |                            | power mgmt board                        |                       | HV aux       | JameCo 236321           |                    |         |         |         |
| A:                                                            | 1: N/C                                       | 3: GND                     | 5: RST                                  | 7: LTCH               | 9: T2        | 11: T1                  | 13: ABS            | 15: GND | 17: P5V | 19: +5V |
|                                                               | 2: N/C                                       | 4: N/C                     | 6: FLT                                  | 8: EBS                | 10: T4       | 12: T3                  | 14: ULTC           | 16: WPR | 18: GND | 20: +5V |
| AWC3 encoder for azimuth (P/N: R176H-03600Q-5L10-AT40SP-24MN) |                                              |                            |                                         |                       |              |                         |                    |         |         |         |
|                                                               | A                                            | Description                | B                                       | Comment               | Color        | Size                    | Twist              | Length  |         |         |
| Wires/pins                                                    | 1                                            | A                          | 4                                       |                       | yellow       | all wires #24 AWG       | Z                  | 5.75    |         |         |
|                                                               | 2                                            | B                          | 6                                       |                       | green        |                         | Y                  | 5.75    |         |         |
|                                                               | 3                                            | IND (Z)                    | 8                                       |                       | blue         |                         | X                  | 5.75    |         |         |
|                                                               | 4                                            | N/C                        |                                         |                       |              |                         |                    |         |         |         |
|                                                               | 5                                            | +5V                        | 10                                      |                       | red          |                         | W                  | 5.75    |         |         |
|                                                               | 6                                            | /A                         | 3                                       |                       | brown        |                         | Z                  | 5.75    |         |         |
|                                                               | 7                                            | /B                         | 5                                       |                       | orange       |                         | Y                  | 5.75    |         |         |
|                                                               | 8                                            | /IND (Z)                   | 7                                       |                       | white        |                         | X                  | 5.75    |         |         |
|                                                               | 9                                            | COMMON                     | 9                                       |                       | black        |                         | W                  | 5.75    |         |         |
| Ends                                                          | A 56-106-010                                 |                            | filtered Dsub 9 socket on control box   |                       |              | connects to AWE12A      |                    |         |         |         |
|                                                               | B WM2522-ND/WM62510-ND                       |                            | 2x5 0.100" on ucntlr card               |                       |              | (22-55-2101/16-02-0102) |                    |         |         |         |
|                                                               |                                              | 1: GND                     | 3: /Z (IND)                             | 5: /B                 | 7: /A        | 9: NC                   |                    |         |         |         |
|                                                               |                                              | 2: +5V                     | 4: Z (IND)                              | 6: B                  | 8: A         | 10: NC                  | in-line w/ battery |         |         |         |

| AWC4 encoder for elevation (P/N: 9220S03600D5L10D99SP04EA) |   |                                             |    |                                                           |             |                    |                         |        |    |
|------------------------------------------------------------|---|---------------------------------------------|----|-----------------------------------------------------------|-------------|--------------------|-------------------------|--------|----|
|                                                            | A | Description                                 | B  | Comment                                                   | Color       | Size               | Twist                   | Length |    |
| Wires/pins                                                 | 1 | A                                           | 4  |                                                           | yellow      | all wires #24 AWG  | Z                       | 5.75   |    |
|                                                            | 2 | B                                           | 6  |                                                           | green       |                    | Y                       | 5.75   |    |
|                                                            | 3 | IND (Z)                                     | 8  |                                                           | blue        |                    | X                       | 5.75   |    |
|                                                            | 4 | N/C                                         |    |                                                           |             |                    |                         |        |    |
|                                                            | 5 | +5V                                         | 10 |                                                           | red         |                    | W                       | 5.75   |    |
|                                                            | 6 | /A                                          | 3  |                                                           | brown       |                    | Z                       | 5.75   |    |
|                                                            | 7 | /B                                          | 5  |                                                           | orange      |                    | Y                       | 5.75   |    |
|                                                            | 8 | /IND (Z)                                    | 7  |                                                           | white       |                    | X                       | 5.75   |    |
|                                                            | 9 | COMMON                                      | 9  |                                                           | black       |                    | W                       | 5.75   |    |
| Ends                                                       | A | 56-106-010                                  |    |                                                           |             | connects to AWE13A |                         |        |    |
|                                                            | B | WM2522-ND/WM62510-ND                        |    | 2x5 0.100" on ucntlr card                                 |             |                    | (22-55-2101/16-02-0102) |        |    |
|                                                            |   | <div>1: GND3: /Z (IND)5: /B7: /A9: NC</div> |    |                                                           |             |                    |                         |        |    |
|                                                            |   | <div>2: +5V4: Z (IND)6: B8: A10: NC</div>   |    |                                                           |             |                    |                         |        |    |
| AWC5 battery                                               |   |                                             |    |                                                           |             |                    |                         |        |    |
|                                                            | A | Description                                 | B  | Comment                                                   | Color       | Size               | Twist                   | Length |    |
| Wires/pins                                                 | 1 | +12V                                        | 1  |                                                           | red         | 18                 |                         | 4      |    |
|                                                            | 2 | +12V                                        | 1  |                                                           | red         | same wire          |                         |        |    |
|                                                            | 3 | N/C                                         |    |                                                           |             |                    |                         |        |    |
|                                                            | 4 | GND                                         | 2  |                                                           | black       | 18                 |                         | 4      |    |
|                                                            | 5 | GND                                         | 2  |                                                           | black       | same wire          |                         |        |    |
|                                                            | 6 | +12V                                        | 1  |                                                           | red         | same wire          |                         |        |    |
|                                                            | 7 | N/C                                         |    |                                                           |             |                    |                         |        |    |
|                                                            | 8 | N/C                                         |    |                                                           |             |                    |                         |        |    |
|                                                            | 9 | GND                                         | 2  |                                                           | black       | same wire          |                         |        |    |
| Ends                                                       | A | 56-101-010                                  |    | filtered Dsub 9 pin on control box                        |             |                    | connects to AWE14A      |        |    |
|                                                            | B | 770849-2/770522-1                           |    |                                                           |             |                    |                         |        |    |
| AWC6                                                       |   |                                             |    |                                                           |             |                    |                         |        |    |
| standard CAT5 cable                                        |   | Rj-45                                       |    |                                                           |             |                    |                         |        | 12 |
|                                                            |   |                                             |    |                                                           |             |                    |                         |        |    |
| AWC7 9V to media converter from power mgmt board           |   |                                             |    |                                                           |             |                    |                         |        |    |
|                                                            | A | Description                                 | B  | Comment                                                   | Color       | Size               | Twist                   | Length |    |
| Wires                                                      | 1 | +9V                                         | 1  |                                                           | black/white | 20                 |                         | 13     |    |
|                                                            | 2 | Common                                      | 2  |                                                           | black       | 20                 |                         | 13     |    |
| Ends                                                       | A | round                                       |    |                                                           |             |                    |                         |        |    |
|                                                            | B | 770849-2/770522-1                           |    | This connector numbering is opposite to 0.100" connectors |             |                    |                         |        |    |
|                                                            |   |                                             |    |                                                           |             |                    |                         |        |    |
| AWC8 24 V from power supply to power mgmt board            |   |                                             |    |                                                           |             |                    |                         |        |    |
|                                                            | A | Description                                 | B  | Comment                                                   | Color       | Size               | Twist                   | Length |    |
| Wires                                                      | 1 | +24V                                        | 1  |                                                           | red         | 18                 |                         | 8.25   |    |
|                                                            | 2 | Common                                      | 2  |                                                           | black       | 18                 |                         | 8.25   |    |
| Ends                                                       | A | spade fork #6x18                            |    |                                                           |             |                    |                         |        |    |
|                                                            | B | 770849-2/770522-1                           |    |                                                           |             |                    |                         |        |    |

| AWC95V to SBC                                                               |   |                        |                                      |                       |                         |      |       |        |
|-----------------------------------------------------------------------------|---|------------------------|--------------------------------------|-----------------------|-------------------------|------|-------|--------|
|                                                                             | A | Description            | B                                    | Comment               | Color                   | Size | Twist | Length |
| Wires                                                                       | 1 | +5V                    | 1                                    |                       | red                     | 18   |       | 7.5    |
|                                                                             | 2 | Common                 | 2                                    |                       | black                   | 18   |       | 7.5    |
| Ends                                                                        | A | 15-24-4048             |                                      |                       |                         |      |       |        |
|                                                                             | B | 770849-2/770522-1      |                                      |                       |                         |      |       |        |
| AWC1012 VDC from Pwer Mgment to uCntrlr board                               |   |                        |                                      |                       |                         |      |       |        |
|                                                                             | A | Description            | B                                    | Comment               | Color                   | Size | Twist | Length |
| Wires                                                                       | 1 | +12V                   | 1                                    |                       | red                     | 18   |       | 4.5    |
|                                                                             | 2 | Common                 | 2                                    |                       | black                   | 18   |       | 4.5    |
| Ends                                                                        | A | 770849-2/770522-1      |                                      |                       |                         |      |       |        |
|                                                                             | B | 770849-2/770522-1      |                                      | 0.156" pitch          |                         |      |       |        |
| AWC11SBC reset (goes between the uCntrlr board and SBC)                     |   |                        |                                      |                       |                         |      |       |        |
|                                                                             | A | Description            | B                                    | Comment               | Color                   | Size | Twist | Length |
| Wires                                                                       | 1 | power                  | 1                                    |                       | green                   | 24   | Z     | 9      |
|                                                                             | 2 | ground                 | 2                                    |                       | black                   | 24   | Z     | 9      |
|                                                                             | 3 | reset                  | 3                                    |                       | red                     | 24   | Z     | 9      |
|                                                                             | 4 | N/C                    | 4                                    | pin 1 on inboard side |                         |      |       |        |
|                                                                             | 5 | N/C                    | 5                                    |                       |                         |      |       |        |
| Ends                                                                        | A | WM2803-ND/WM62510-ND   | 1x5 100mil connector on ucntrlr card |                       | (50-57-9005/16-02-0102) |      |       |        |
|                                                                             | B | WM2803-ND/WM62510-ND   | 1x5 100mil connector on SBC          |                       | (50-57-9005/16-02-0102) |      |       |        |
| AWC12electronics on/off (between uCntrlr board and power management board)  |   |                        |                                      |                       |                         |      |       |        |
|                                                                             | A | Description            | B                                    | Comment               | Color                   | Size | Twist | Length |
| Wires                                                                       | 1 | electronics on/off     | 1                                    |                       | green                   | 24   | Z     | 6.5    |
|                                                                             | 2 | ground                 | 2                                    |                       | black                   | 24   | Z     | 6.5    |
|                                                                             | 3 | battery voltage /3     | 3                                    |                       | yellow                  | 24   | Z     | 6.5    |
| Ends                                                                        | A | 1x3 0.100" unpolarized |                                      |                       |                         |      |       |        |
|                                                                             | B | 1x3 0.100" unpolarized |                                      |                       |                         |      |       |        |
| AWC13short cat 5 crossover cable between SBC (second LAN) and uCntrlr board |   |                        |                                      |                       |                         |      |       |        |
| crossover CAT5 cable                                                        |   |                        |                                      |                       |                         |      |       | 6"     |

**Alidade wire harnessing: drive box**

Reference ELMCOoverview.vsd

7-Mar-06

**AWD1 3-wire power cable from cover to rectifier**

|       | A | Description                                       | B                              | Color | Size                               | Twist | Length |
|-------|---|---------------------------------------------------|--------------------------------|-------|------------------------------------|-------|--------|
| Wires | 1 | 100VAC (L)                                        |                                | white | 18AWG                              |       | 13.75" |
|       | 2 | Return (N)                                        |                                | black |                                    |       | 13.75" |
|       | 3 | Ground                                            |                                | green |                                    |       | 6.5"   |
| Ends  | A | 0.25 quick disconnect solder terminals on IEC-320 |                                |       |                                    |       |        |
|       | B | bare wire                                         | screw terminals on drive board |       | (soldered butt splice heat shrink) |       |        |

**AWD2 15-wire cable from drive-box top to daughter board/drive board**

|            | A  | Description      | B,C,D                                                         | Color  | Size             | Twist | Length |
|------------|----|------------------|---------------------------------------------------------------|--------|------------------|-------|--------|
| Wires/pins | 1  | az brakes +24V   | C                                                             | Brown  | all wires #22AWG |       | 4.625  |
|            | 2  | Ground           | C                                                             | Black  |                  |       | 4.625  |
|            | 3  | 24V aux          | C                                                             | Red    |                  |       | 4.625  |
|            | 4  | Ground           | D                                                             | Black  |                  |       | 9      |
|            | 5  | drive enable +   | B1                                                            | Orange |                  |       | 12     |
|            | 6  | Ground           | D                                                             | Black  |                  |       | 9      |
|            | 7  | Az cw limit      | B3                                                            | Yellow |                  |       | 12     |
|            | 8  | Az limit common  | B4                                                            | Green  |                  |       | 12     |
|            | 9  | el brakes +24V   | C                                                             | Blue   |                  |       | 4.625  |
|            | 10 | Ground           | C                                                             | Black  |                  |       | 4.625  |
|            | 11 | 24V aux          | C                                                             | Purple |                  |       | 4.625  |
|            | 12 | Ground           | D                                                             | Black  |                  |       | 9      |
|            | 13 | drive enable -   | B2                                                            | Grey   |                  |       | 12     |
|            | 14 | Ground           | D                                                             | Black  |                  |       | 9      |
|            | 15 | Az ccw limit     | B5                                                            | White  |                  |       | 12     |
| Ends       | A  | 56-111-010       | filtered Dsub15P on driver box top connects to AWE6B          |        |                  |       |        |
|            | B  | WM2803-ND/WM625  | 1x5 100mil connector on daughter card (50-57-9005/16-02-0102) |        |                  |       |        |
|            | C  | bare wire/solder | screw terminal on driver board                                |        |                  |       |        |
|            | D  | gnd lug          | round #8                                                      |        |                  |       |        |

**AWD3 temperature sensor cable from drive board to daughter card**

|  | A | Description | B                    | Color  | Size          | Twist | Length |
|--|---|-------------|----------------------|--------|---------------|-------|--------|
|  | 1 | GND         | 3                    | black  |               |       | 5      |
|  | 2 | T           | 2                    | yellow |               |       | 5      |
|  | 3 | +5          | 1                    | red    |               |       | 5      |
|  | A |             | 50-57-9003           |        | (on daughter) |       |        |
|  | B |             | 22-01-2037/08-52-123 |        | (on driver)   |       |        |

**AWD4 temperature sensor cable from heatsink to daughter card**

|  | A | Description | B                    | Color  | Size          | Twist | Length |
|--|---|-------------|----------------------|--------|---------------|-------|--------|
|  | 1 | GND         |                      | black  |               | 24    | 19     |
|  | 2 | T           |                      | yellow |               | 24    | 19     |
|  | 3 | +5          |                      | red    |               | 24    | 19     |
|  | A |             | 50-57-9003           |        | (on daughter) |       |        |
|  | B |             | soldered/heat shrink |        | LM35          |       |        |

**AWD5 temperature sensor cable from airflow to daughter card**

|  | A | Description | B                    | Color  | Size          | Twist | Length |
|--|---|-------------|----------------------|--------|---------------|-------|--------|
|  | 1 | GND         |                      | black  |               | 24    | 16.75  |
|  | 2 | T           |                      | yellow |               | 24    | 16.75  |
|  | 3 | +5          |                      | red    |               | 24    | 16.75  |
|  | A |             | 50-57-9003           |        | (on daughter) |       |        |
|  | B |             | soldered/heat shrink |        | LM35          |       |        |

**AWD6 rectifier to capacitor 1**

|  | A | Description | B,C   | Color | Size | Twist | Length |
|--|---|-------------|-------|-------|------|-------|--------|
|  | 1 |             | B,C 1 | Red   | 18   |       | 7"     |
|  | 2 |             | B,C 2 | Black | 18   |       | 7"     |
|  | A |             |       |       |      |       |        |
|  | B |             |       |       |      |       |        |
|  | C |             |       |       |      |       |        |

**AWD7 capacitor 1 to capacitor 2**

|  | A | Description               | B | Color   | Size | Twist | Length |
|--|---|---------------------------|---|---------|------|-------|--------|
|  | 1 | +                         |   | 1 Red   | 18   |       | 8.5"   |
|  | 2 | -                         |   | 2 Black | 18   |       | 8.5"   |
|  | A | #8 ring terminal + solder |   |         |      |       |        |
|  | B | #8 ring terminal + solder |   |         |      |       |        |

**AWD8 resistor (6) to drive board**

|  | A | Description | B | Color | Size | Twist | Length |
|--|---|-------------|---|-------|------|-------|--------|
|  | 1 |             |   | Red   | 18   |       | 9.5"   |
|  | 2 |             |   | Black | 18   |       | 7.5"   |
|  | A |             |   |       |      |       |        |
|  | B |             |   |       |      |       |        |

**AWD9 resistor to rectifier**

|  | A | Description | B | Color   | Size | Twist | Length |
|--|---|-------------|---|---------|------|-------|--------|
|  | 1 |             |   | 1 Red   | #22  |       | 4"     |
|  | 2 |             |   | 2 Black | #22  |       | 4"     |